

2520030395

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1. Develop a C program that demonstrates how a Linux operating system executes a command entered by a user that 1. Accept a Linux command as input. 2. Create a child process using fork(). 3. Execute the command in the child process using an appropriate exec() system call. 4. Allow the parent process to wait for the child using wait (). 5. Display the Process ID (PID) of both parent and child processes.

Using Linux terminal commands (uname, lscpu, lsblk, ps, top), investigate the relationship between hardware resources and operating system services. Prepare a report explaining how the OS abstracts CPU, memory, storage, and I/O devices.

```
#include <stdio.h>
#include <stdlib.h>
#include <unistd.h>
#include <sys/wait.h>

int main()
{
    char cmd[100];

    printf("Enter Linux command: ");
    scanf("%s", cmd);
    pid_t pid = fork();
    if(pid == 0)
    {
        printf("Child PID : %d\n", getpid());
        execlp(cmd, cmd, NULL);
        perror("Execution Failed");
    }
    else if(pid > 0)
    {
        printf("Parent PID : %d\n", getpid());
        wait(NULL);
        printf("Child process completed.\n");
    }
    else
    {
        printf("Fork Failed\n");
    }
    return 0;
}
```

Linux command execution using fork(), exec(), wait()

```
monish_putti@monhh: ~  
monish_putti@monhh:~$ nano practical1.c  
monish_putti@monhh:~$ gcc practical1.c -o practical1  
monish_putti@monhh:~$ ./practical1  
Enter Linux command: uname /a  
Parent PID : 687  
Child PID : 690  
Linux  
Child process completed.  
monish_putti@monhh:~$
```

```
monish_putti@monhh: ~  
monish_putti@monhh:~$ ./practical1  
Enter Linux command: lscpu  
Parent PID : 704  
Child PID : 706  
Architecture:          x86_64  
CPU op-mode(s):        32-bit, 64-bit  
Address sizes:          46 bits physical, 48 bits virtual  
Byte Order:             Little Endian  
CPU(s):                 18  
On-line CPU(s) list:    0-17  
Vendor ID:              GenuineIntel  
Model name:             Intel(R) Core(TM) Ultra 5 125H  
CPU family:             6  
Model:                  170  
Thread(s) per core:     2  
Core(s) per socket:     9  
Socket(s):              1  
Stepping:               4  
BogoMIPS:               5990.40  
Flags:                  fpu vme de pse tsc msr pae mce cx8 apic sep mtrr pge mca cmov pat pse36 clflush mmx fxsr sse s  
se2 ss ht syscall nx pdpe1gb rdtscp lm constant_tsc rep_good nopl xtopology tsc_reliable nonst  
op_tsc cpuid tsc_known_freq pni pclmulqdq vmx ssse3 fma cx16 pcid sse4_1 sse4_2 x2apic movbe p  
opcnt tsc_deadline_timer aes xsave avx f16c rdrand hypervisor lahf_lm abm 3dnowprefetch ssbd i  
brs ibpb stibp ibrs_enhanced tpr_shadow ept vpid ept_ad fsgsbase tsc_adjust bmi1 avx2 smep bmi  
2 erms invpcid rdseed adx smap clflushopt clwb sha_ni xsaveopt xsavec xgetbv1 xsaves avx_vnni  
vnmi umip waitpkg gfni vaes vpclmulqdq rdpid movdiri movdir64b fsrm md_clear serialize flush_l  
1d arch_capabilities  
  
Virtualization features:  
Virtualization:         VT-x  
Hypervisor vendor:      Microsoft
```

```

monish_putti@monhh: ~
Child process completed.
monish_putti@monhh:~$ ./practical1
Enter Linux command: lsblk
Parent PID : 720
Child PID : 721
NAME MAJ:MIN RM   SIZE RO TYPE MOUNTPOINTS
sda   8:0    0 388.4M  1 disk
sdb   8:16   0  186M  1 disk
sdc   8:32   0    2G   0 disk [SWAP]
sdd   8:48   0    1T   0 disk /mnt/wslg/distro
/

Child process completed.
monish_putti@monhh:~$ ./practical1
Enter Linux command: ps
Parent PID : 731
Child PID : 734
  PID TTY          TIME CMD
  628 pts/0        00:00:00 bash
   731 pts/0        00:00:00 practical1
   734 pts/0        00:00:00 ps

Child process completed.
monish_putti@monhh:~$ ./practical1
Enter Linux command: top
Parent PID : 744
Child PID : 782
top - 09:25:55 up 6 min,  1 user,  load average: 0.00, 0.01, 0.00
Tasks:  27 total,   1 running, 26 sleeping,   0 stopped,   0 zombie
%Cpu(s):  0.0 us,  0.0 sy,  0.0 ni,100.0 id,  0.0 wa,  0.0 hi,  0.0 si,  0.0 st
MiB Mem : 7651.7 total, 6677.8 free,  582.4 used,  543.8 buff/cache
MiB Swap: 2048.0 total, 2048.0 free,   0.0 used, 7069.3 avail Mem

monish_putti@monhh: ~
%Cpu(s):  0.0 us,  0.0 sy,  0.0 ni,100.0 id,  0.0 wa,  0.0 hi,  0.0 si,  0.0 st
MiB Mem : 7651.7 total, 6677.8 free,  582.4 used,  543.8 buff/cache
MiB Swap: 2048.0 total, 2048.0 free,   0.0 used, 7069.3 avail Mem

  PID USER      PR  NI  VIRT  RES  SHR  S  %CPU  %MEM    TIME+  COMMAND
  782 monish_+  20   0   8148   5760  3744  R   0.3   0.1   0:00.02 top
    1 root      20   0  24364 14248 10936  S   0.0   0.2   0:00.83 systemd
    2 root      20   0   3072   1728  1728  S   0.0   0.0   0:00.01 init-systemd(Ub
    8 root      20   0   3072   1872  1872  S   0.0   0.0   0:00.00 init
   48 root      19  -1  50396 15984 14976  S   0.0   0.2   0:00.20 systemd-journal
   85 systemd+  20   0  22400 13968 11520  S   0.0   0.2   0:00.10 systemd-resolve
   93 root      20   0  35288 11952  8928  S   0.0   0.2   0:00.49 systemd-udev
  114 root      20   0   2880   1728  1728  S   0.0   0.0   0:00.05 chronyd-starter
  121 root      20   0   4464   2736  2592  S   0.0   0.0   0:00.01 cron
  126 message+  20   0   8780   4320  4176  S   0.0   0.1   0:00.08 dbus-daemon
  158 root      20   0  44080 27936 13536  S   0.0   0.4   0:00.34 networkd-dispat
  180 root      20   0  18424  8352  7344  S   0.0   0.1   0:00.12 systemd-logind
  186 root      20   0 1866024 13392 11376  S   0.0   0.2   0:00.15 wsl-pro-service
  207 syslog    20   0 220532  5040  4176  S   0.0   0.1   0:00.09 rsyslogd
  248 _chrony     20   0  24012 10512  6480  S   0.0   0.1   0:00.11 chronyd
  250 _chrony     20   0  12080  2332  1728  S   0.0   0.0   0:00.00 chronyd
  260 root      20   0   5528   2736  2592  S   0.0   0.0   0:00.01 agetty
  311 root      20   0   5484   2592  2448  S   0.0   0.0   0:00.01 agetty
  313 root      20   0 123444 32112 17568  S   0.0   0.4   0:00.24 unattended-upgr
  412 root      20   0   8632   5328  4608  S   0.0   0.1   0:00.01 login
  463 monish_+  20   0  22328 12528 10512  S   0.0   0.2   0:00.14 systemd
  465 monish_+  20   0  22896  4012  2016  S   0.0   0.1   0:00.00 (sd-pam)
  493 monish_+  20   0   6128   5184  3744  S   0.0   0.1   0:00.04 bash

Child process completed.
monish_putti@monhh:~$

```

2. Develop a C program that uses the system calls `open()`, `read()`, `write()`, and `close()` to copy the contents of one file to another. Explain how control transitions between user space and kernel space during execution.

Use the `strace` utility to trace all system calls generated by the command `cat sample.txt`. Analyze the sequence of system calls and identify the kernel services involved.

